

AJCR Human Mathematician Review Packet

Snapshot: 2026-08-24 Workspace ledger: 16de23f3d5

Purpose. This packet is a compact, auditable briefing for a human mathematician. It is not a completion claim. The question is whether the current finite-descent route can produce an unconditional original-field Picard-zero representer, or whether the route must change.

Executive diagnosis

The rank-one and separably-closed portions are genuine and kernel-clean. The arbitrary-field capstone is absent. The two decisive missing bridges are:

1. a binder-free finite-stage `RepresentableBy`/Yoneda equivalence for the glued carrier;
2. exact-carrier projectivity or `FiniteInAffine` control of the finite Galois orbit.

The compile chain needed to test these bridges is also uncertified, so many sessions have produced prerequisites and conditional consumers without moving the headline theorem.

Acceptance boundary

Edge	Status	Meaning for review
<code>canonicalRankOneAbelIso</code>	Proved producer	Rank-one divisor and Picard loci are identified over the base field; narrow axioms are <code>propext</code> , <code>Classical.choice</code> , <code>Quot.sound</code> .
<code>pic0_sepClosed_representableBy</code>	Proved, sep-closed only	Produces the exact <code>J/rep</code> pair under <code>[IsSepClosed k]</code> .
Finite-stage Galois wrappers	Conditional consumer	Bind an explicit <code>rep</code> and orbit/projectivity hypotheses; they are not producers.
Glue package chain	Compile frontier	<code>GluePackage -> GluingDiagramIso -> PreSnd -> OverlapIsoSnd -> GluedComparison</code> lacks native artifacts.
<code>pic0_representableBy</code>	Missing producer	No unconditional original-field <code>RepresentableBy</code> theorem exists at the current checkpoint.
<code>Challenge.Jacobian</code>	Missing handoff	No original-field <code>JacobianData</code> consumer is wired in; <code>sorryAx</code> remains.

Critical path

```
canonicalRankOneAbelIso
  -> pic0_sepClosed_representableBy
  -> picRepDatumSepClosed -> jacobianDataSepClosed

finite-stage package
  -> GluePackage -> GluingDiagramIso -> PreSnd
  -> OverlapIsoSnd -> GluedComparison
  -> binder-free RepresentableBy P.gluedOver
  -> exact-carrier projectivity / FiniteInAffine
  -> original-field pic0_representableBy
  -> PicRepDatum -> JacobianData -> Challenge.Jacobian
```

Evidence and blockers

- `GluePackage.lean` fails before its proof body at about 7.1 GB RSS.

- `overlapBaseChangeIso_hom_iota` is the first expensive declaration in the coherent cone; prior checks reached about 610 seconds.
- Run 0154 produced no `GluePackage.olean` or `GluingDiagramIso.olean`; downstream checks cannot proceed.
- The mathematical blockers are universal Yoneda descent, exact-carrier orbit geometry, and original-field assembly.
- The strategy audit classified one implementation interval as 266 commits but 0 acceptance edges, 82 consumed pre-requisites, 10 conditional consumers, 11 compile/performance splits, and 163 metadata/churn.

Questions for the reviewer

1. Is the finite-stage atlas sufficient for a natural `homEquiv`, or should a standard Picard representability theorem be used?
2. Can projectivity or `FiniteInAffine` be proved for the exact glued carrier?
3. Does the current `Challenge.lean` statement boundary match the available original-field data?
4. What single theorem should be the next acceptance milestone?

Reading order

This packet, then the “Acceptance review” section of `MainProjects/Algebraic-Jacobian-Challenge-Rebuild/README.md`; next read the reports in `.archon-horizon/runs/0152`, `0153`, `0154`, and `0156`; finally inspect `Pic0CriticalPath.lean` and the two reviewer PDFs under `MainProjects/Algebraic-Jacobian-Challenge/informal/`.